

Joseph Flammia

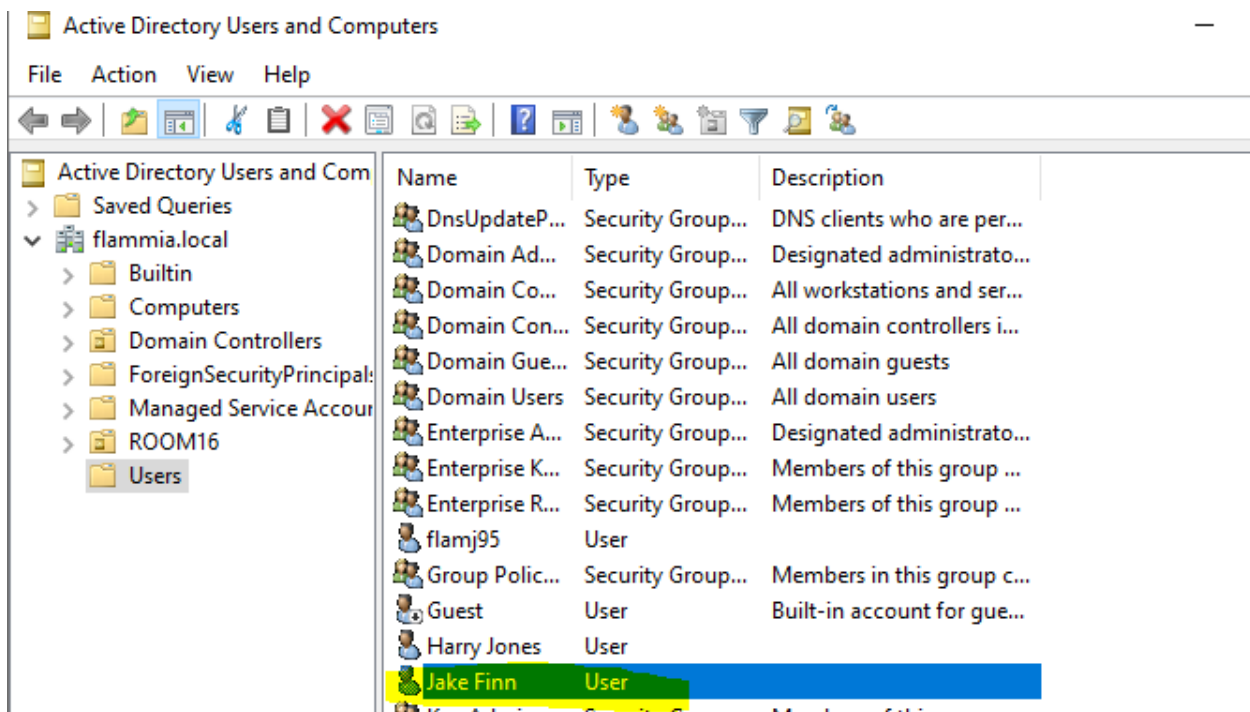
2/19/25

Lab 7

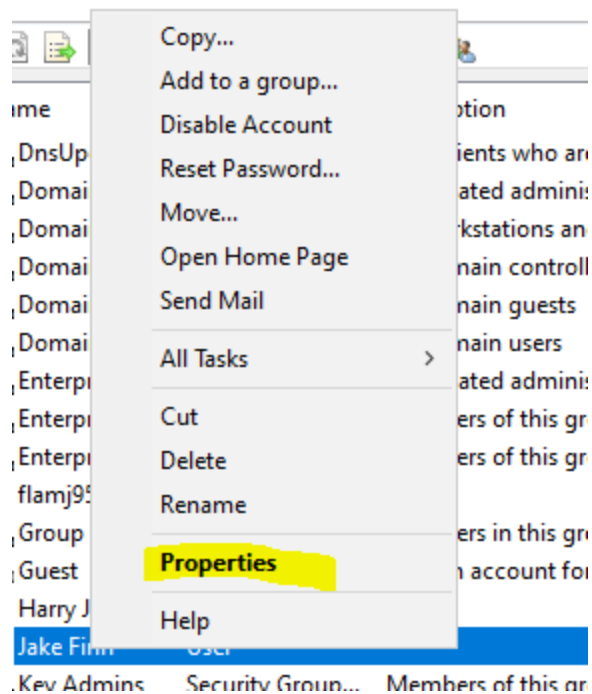
CYB-126

Lab 7A:

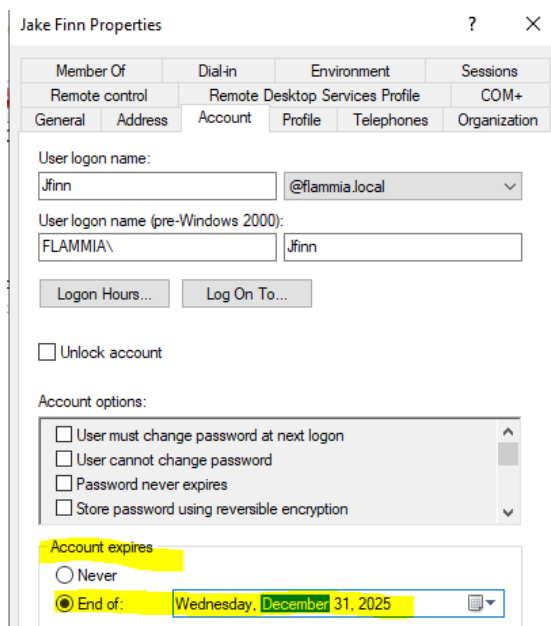
To start off, we will create a new account on our AD server using “Active Directory Users and Computers.” As shown below, I created a new user named “Jake Finn.”



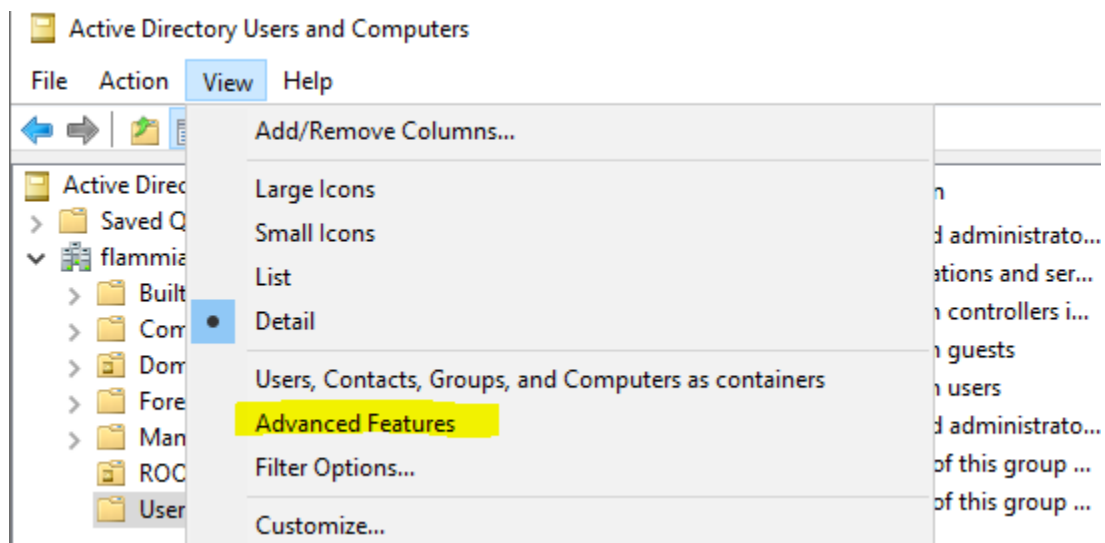
Next, we will set an expiration date for the newly created account. To do this, we will open the users' properties by right clicking on their name.



Once we have the properties opened, we will navigate to the “account” tab. At the bottom of this tab, we can set an expiration date for the account. In the picture below, I set the account expiration date for the end of Wednesday, December 31st of 2025.



We can also use the attribute editor to view the properties of certain users in our domain. To do this, click on the “view” tab in AD Users and Computers. Then, click on “Advanced Features.”



Once the advanced features option has been toggled, we now have access to even more information about users in the domain. Now if we click on the properties of a user, we can see the attribute editor tab. As you can see, the account expiration date is listed below as “1/1/2026”.

Jake Finn Properties ?

Published Certificates	Member Of	Password Replication	Dial-in	Obj...
Security	Environment	Sessions	Remote control	
General	Address	Account	Profile	Telephones
Remote Desktop Services Profile		COM+	Attribute Editor	

Attributes:

Attribute	Value
accountExpires	1/1/2026 12:00:00 AM Eastern Standard Time
accountNameHistory	<not set>
aCSPolicyName	<not set>
adminCount	<not set>
adminDescription	<not set>
adminDisplayName	<not set>
altSecurityIdentities	<not set>
assistant	<not set>
attributeCertificateAttri...	<not set>
audio	<not set>
badPasswordTime	(never)
badPwdCount	0
businessCategory	<not set>
c	<not set>

Edit Filter

We are also able to view attributes of users in our domain via Windows PowerShell on our AD server. In PowerShell, I entered command “get-aduser Jfinn -Properties *.” In the picture below, we can see some properties of a user, like the attribute editor in Users and Computers.

As pictured below, we can see the expiration date for the user “JFinn,” just like in the Users and Computers Application.

```
Administrator: Windows PowerShell
Windows PowerShell
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improvements! https://aka.ms/PSWindows
PS C:\Windows\system32> get-aduser JFinn -Properties *

AccountExpirationDate      : 1/1/2026 12:00:00 AM
AccountExpires             : 134117172000000000
AccountLockoutTime         :
AccountNotDelegated        : False
AllowReversiblePasswordEncryption : False
AuthenticationPolicy       : {}
AuthenticationPolicySilo   : {}
BadLogonCount              : 0
BadPasswordTime            : 0
BadPwdCount                : 0
CannotChangePassword       : False
CanonicalName              : flammia.local/Users/Jake Finn
Certificates               : {}
City                       :
CN                         : Jake Finn
CodePage                   : 0
Company                   : Windows
CompoundIdentitySupported  : {}
Country                   :
CountryCode               : 0
Created                   : 2/19/2025 2:05:53 PM
createTimeStamp            : 2/19/2025 2:05:53 PM
Deleted                   :
```

Here are some other examples of finding properties in the Users and Computers app vs. PowerShell. In the picture below, I set some basic information for the user “Jake Finn,” such as a job title, department, and company. This information can all be found within the “Organization” tab in Users and Computers.

The screenshot shows the 'Jake Finn Properties' dialog box with the 'Organization' tab selected. The 'Job Title' is 'Employee', 'Department' is 'IT', and 'Company' is 'Windows'. The 'Manager' section is empty, and 'Direct reports' is also empty.

Published Certificates	Member Of	Password Replication	Dial-in	Object
Remote Desktop Services Profile		COM+	Attribute Editor	
Security	Environment	Sessions	Remote control	
General	Address	Account	Profile	Telephones
Organization				

Job Title:

Department:

Company:

Manager

Name:

Direct reports:

Now in PowerShell using the command “get-aduser Jfinn -Properties *,” we can see the information that we have just found in the application. As highlighted below, we can see the company, title, and department that the user “JFinn” has currently.

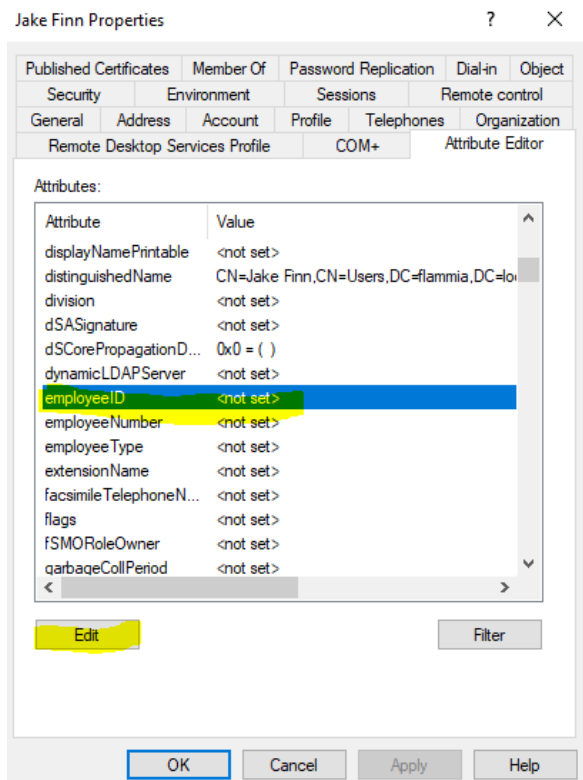
```
PS C:\Users\flamj95> get-aduser JFinn -Properties *

AccountExpirationDate      :
accountExpires             :
AccountLockoutTime         :
AuthenticationPolicy       : {}
AuthenticationPolicySilo   : {}
BadLogonCount              :
CannotChangePassword       : False
CanonicalName              :
Certificates               : {}
City                      :
CN                         : Jake Finn
codePage                   : 0
Company                    : Windows
CompoundIdentitySupported   : {}
Country                   :
countryCode                : 0
Created                   :
Deleted                   :
Department                 : IT
Description                :
```

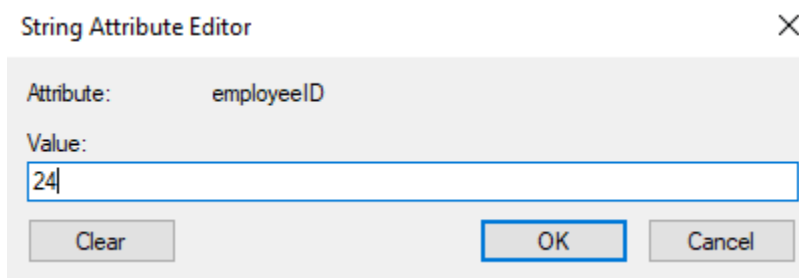
Select Windows PowerShell

```
OtherName :  
PasswordLastSet :  
POBox :  
PostalCode :  
PrimaryGroup : CN=Domain Users,CN=Users,DC=flammia,DC=local  
primaryGroupID : 513  
PrincipalsAllowedToDelegateToAccount : {}  
ProfilePath :  
ProtectedFromAccidentalDeletion : False  
SamAccountName : Jfinn  
sAMAccountType : 805306368  
ScriptPath :  
sDRightsEffective : 4  
ServicePrincipalNames : {}  
SID : S-1-5-21-2271592887-2331845800-14120086-2103  
SIDHistory : {}  
sn : Finn  
State :  
StreetAddress :  
Surname : Finn  
Title : Employee  
userCertificate : {}  
UserPrincipalName : Jfinn@flammia.local
```

Next up, we will add an employee ID to the user Jake Finn. To do this, we will open our users' properties just like before, and navigate to the attribute editor tab. After scrolling down the list of attributes, we can see that the employeeID field is blank. To edit this, all we must do is click the “Edit” button.



Once in the editor, we can now add a value for the “employeeID.” Here, I entered the value as “24”.



We can now see that the user “Jake Finn” has an employee ID of 24 as pictured below.

Jake Finn Properties ? >

Published Certificates	Member Of	Password Replication	Dial-in	Object
Security	Environment	Sessions	Remote control	
General	Address	Account	Profile	Telephones
Remote Desktop Services Profile	COM+	Attribute Editor		

Attributes:

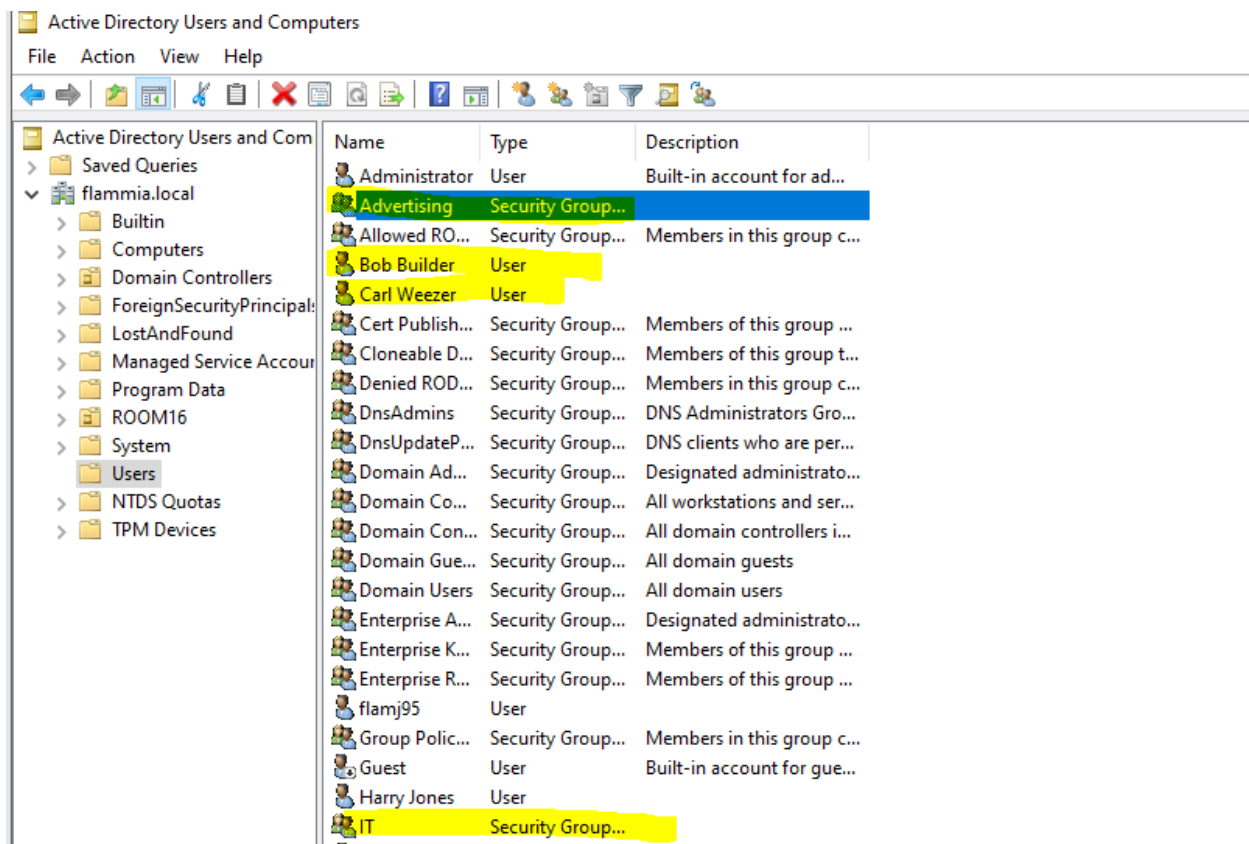
Attribute	Value
displayNamePrintable	<not set>
distinguishedName	CN=Jake Finn,CN=Users,DC=flammia,DC=lo
division	<not set>
dSASignature	<not set>
dSCorePropagationD...	0x0 = ()
dynamicLDAPServer	<not set>
employeeID	24
employeeNumber	<not set>
employeeType	<not set>
extensionName	<not set>
facsimileTelephoneN...	<not set>
flags	<not set>
fSMORoleOwner	<not set>
garbageCollPeriod	<not set>

< >

Edit Filter

Lab 7B:

To start off, we will create two groups and two accounts in AD Users and Computers. We will put one of these accounts in of the groups and the other account in the other group. Here, I created two users named “Bob Builder” and “Carl Weezer.” I then created two groups named “IT” and “Advertising.”



I then added “Bob Builder” to the IT group, and “Carl Weezer” to the Advertising group.

Carl Weezer User

Select Groups

Select this object type:
Groups or Built-in security principals Object Types...

From this location:
flammia.local Locations...

Enter the object names to select (examples):
Advertising Check Names

Advanced... OK Cancel

Bob Builder User

Select Groups

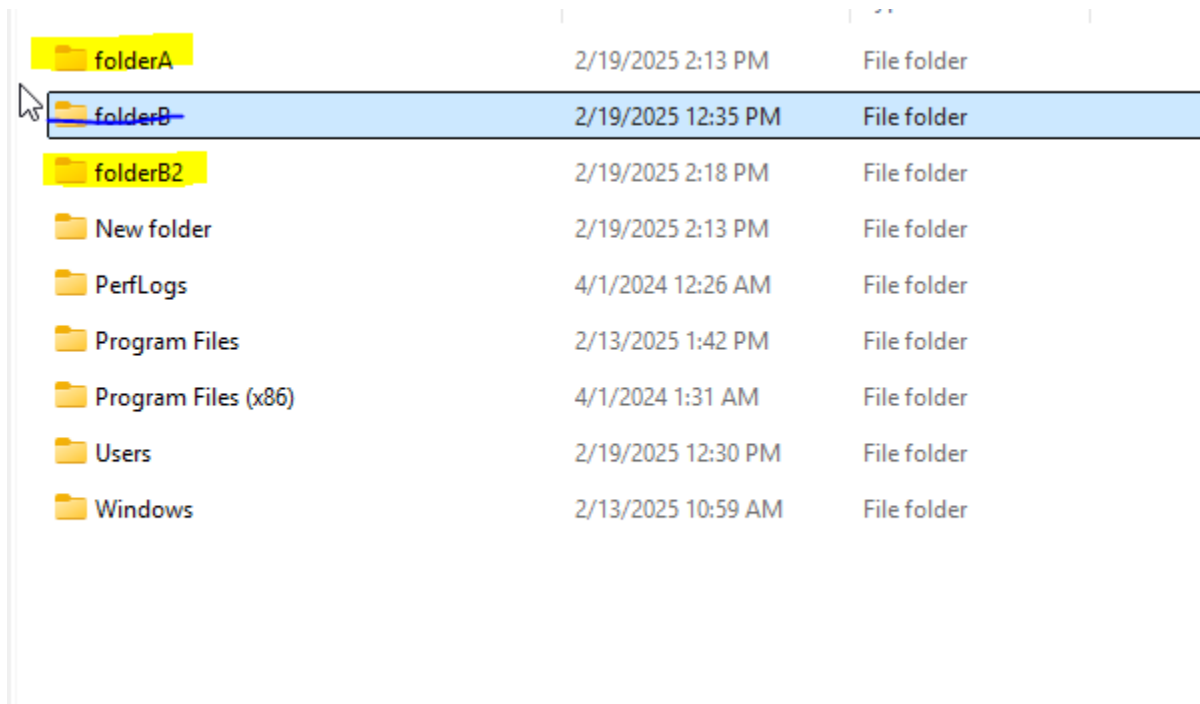
Select this object type:
Groups or Built-in security principals Object Types...

From this location:
flammia.local Locations...

Enter the object names to select (examples):
IT Check Names

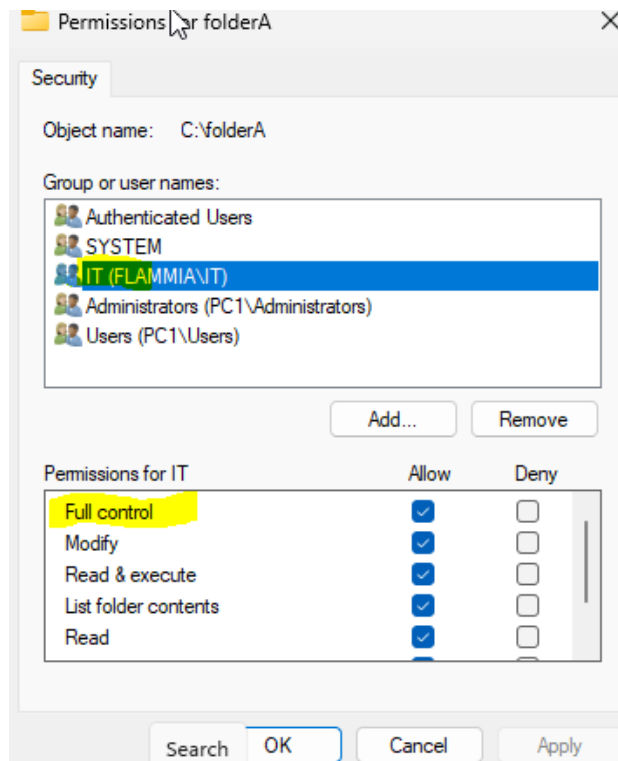
Advanced... OK Cancel

Switching over to my Windows 11 Client, I created two folders on the C Drive named “folderA”, and “folderB2”.

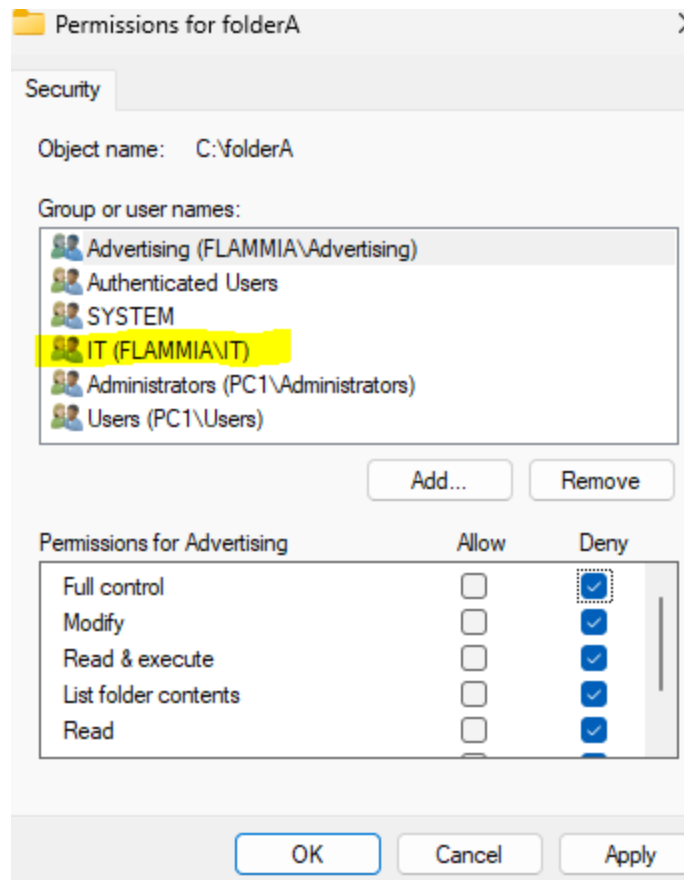


folderA	2/19/2025 2:13 PM	File folder
folderB	2/19/2025 12:35 PM	File folder
folderB2	2/19/2025 2:18 PM	File folder
New folder	2/19/2025 2:13 PM	File folder
PerfLogs	4/1/2024 12:26 AM	File folder
Program Files	2/13/2025 1:42 PM	File folder
Program Files (x86)	4/1/2024 1:31 AM	File folder
Users	2/19/2025 12:30 PM	File folder
Windows	2/13/2025 10:59 AM	File folder

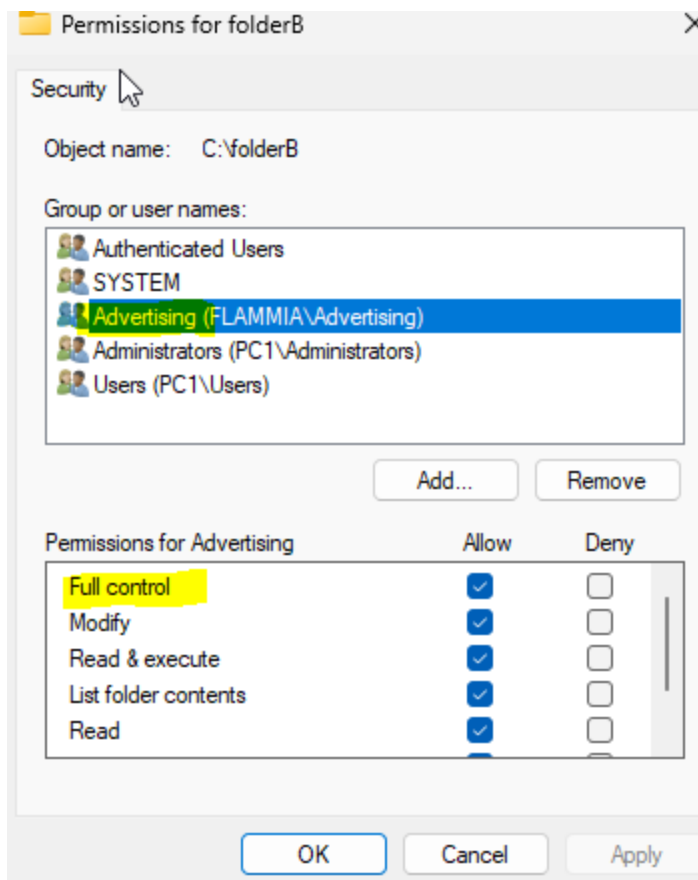
Next, I made it so that the IT group has full control over “folderA,” and Advertising has full control over “folderB2”. I did this by navigating to the “permissions” tab of the folders. As pictured below, I granted IT full control and

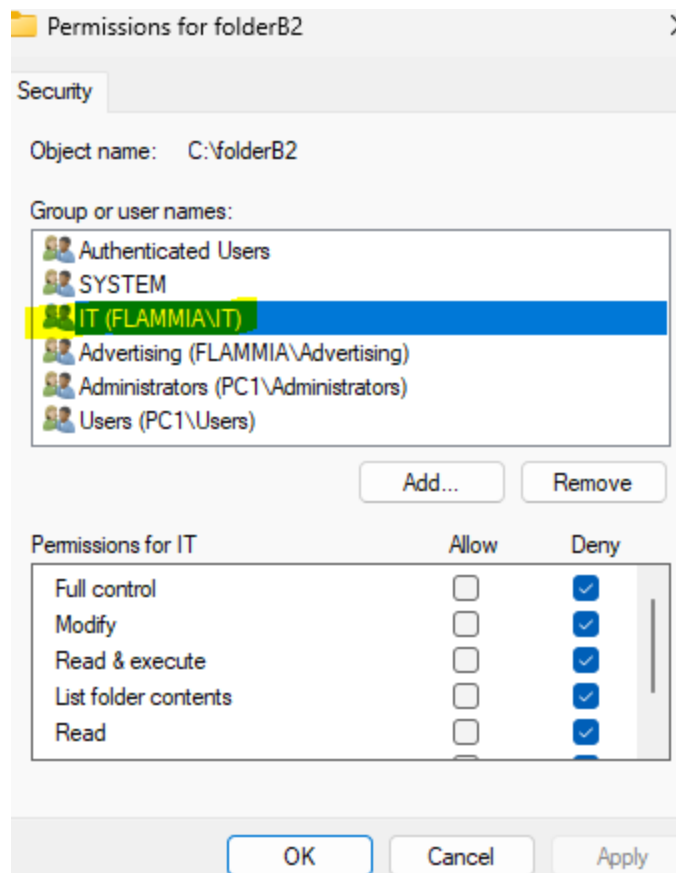


After granting the IT group all permissions for folder A, I removed all permissions from the Advertising Group.



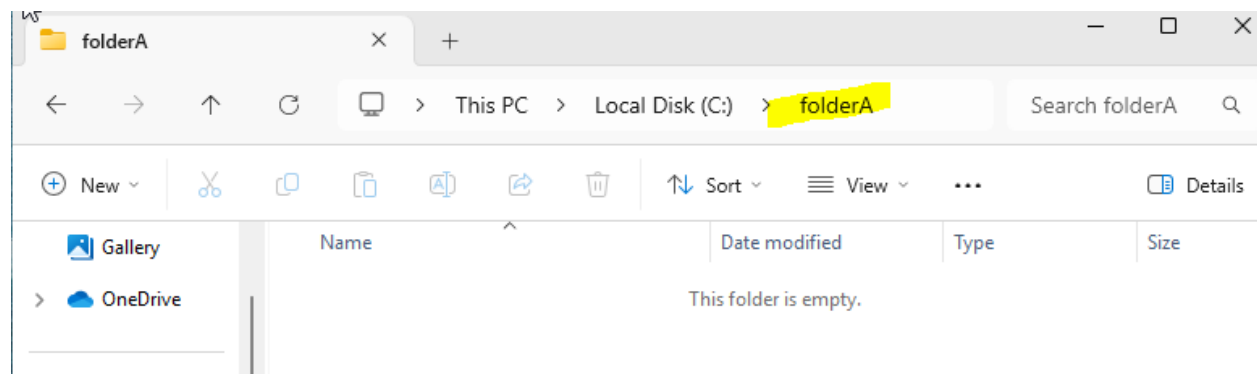
I then did this same process for folderB2, except I gave the Advertising group full control of that folder and removed all permissions for the IT group.

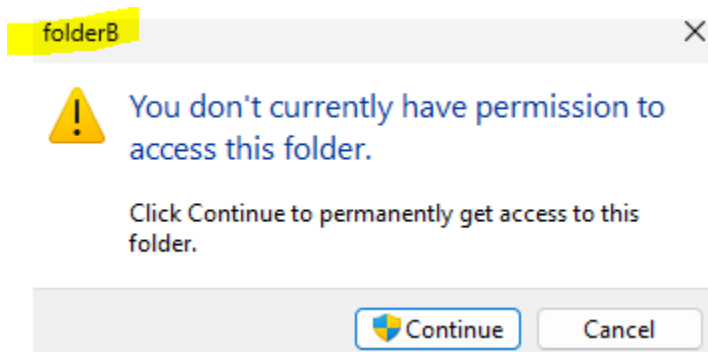




Next, I will log out of the Windows 11 Client and log in with the “BBuilder” account and I will try to access both folders.

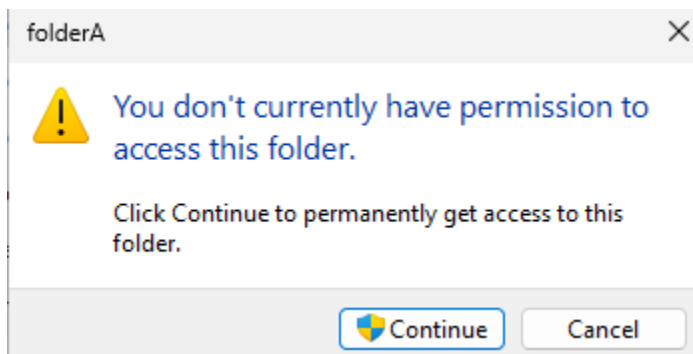
As you can see below, I can access folder A with no issues. However, when trying to access folder B, I am prompted with an error message.

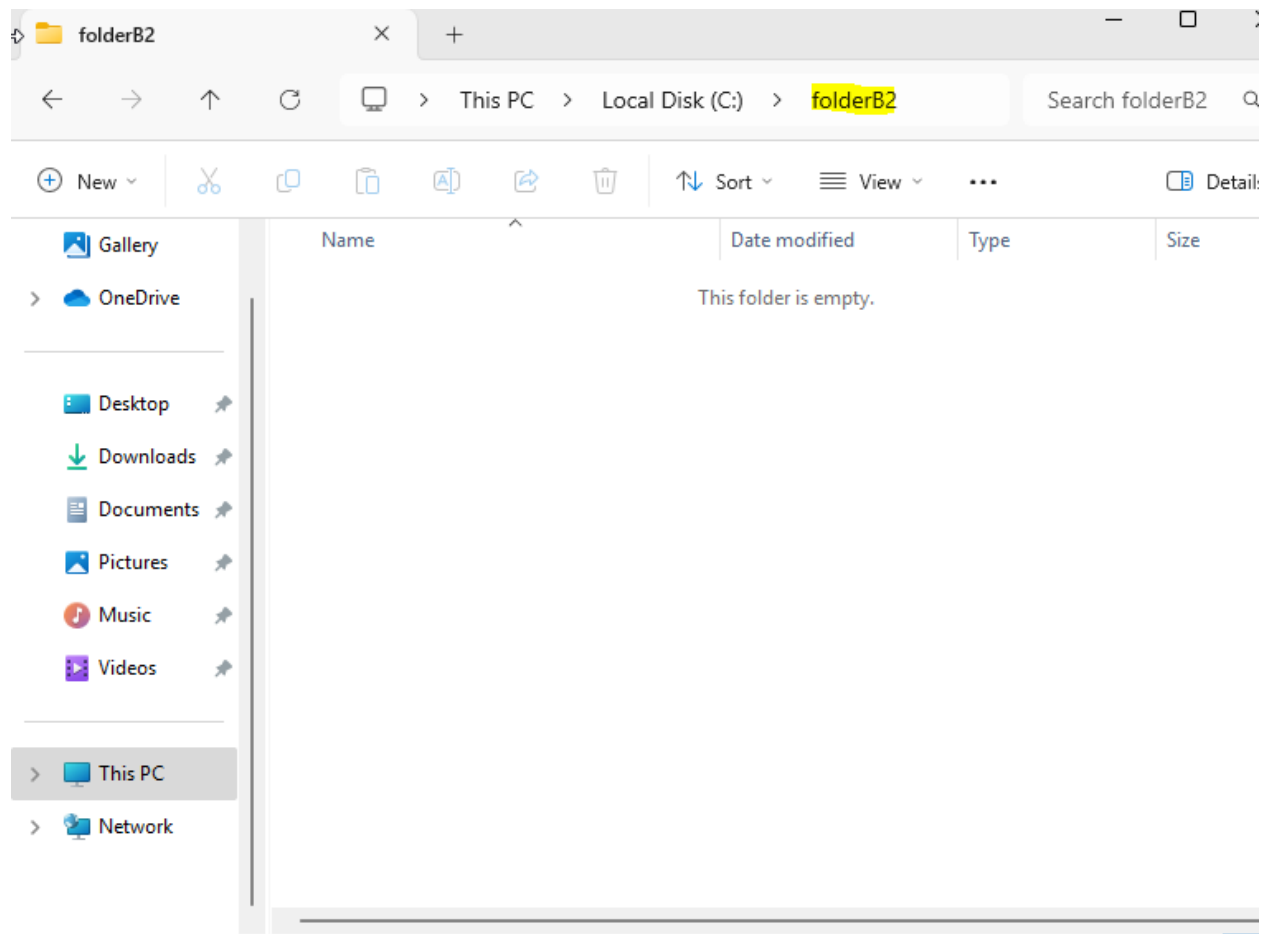




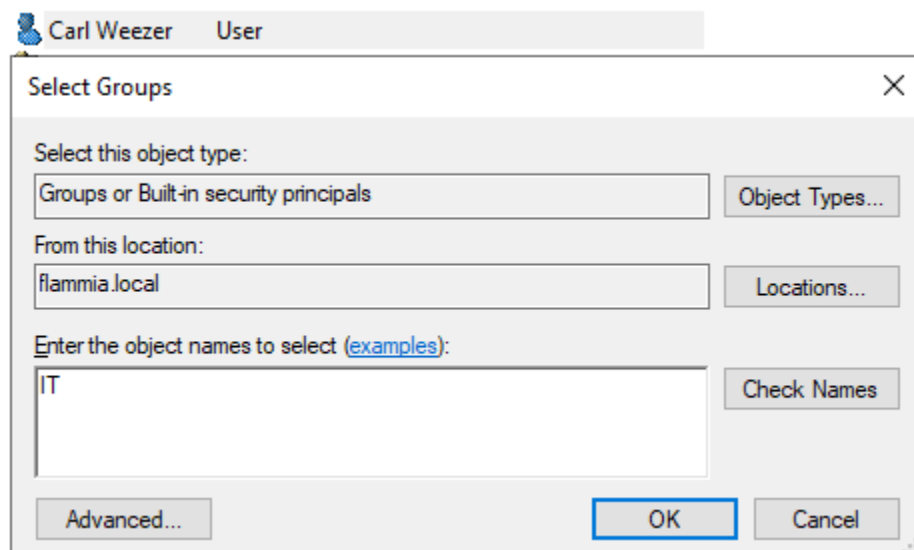
Now, I will login with my second user (CWeezer) and attempt to open both files.

As you can see below, I am denied access to folder A, but I am able to freely open folder B2.

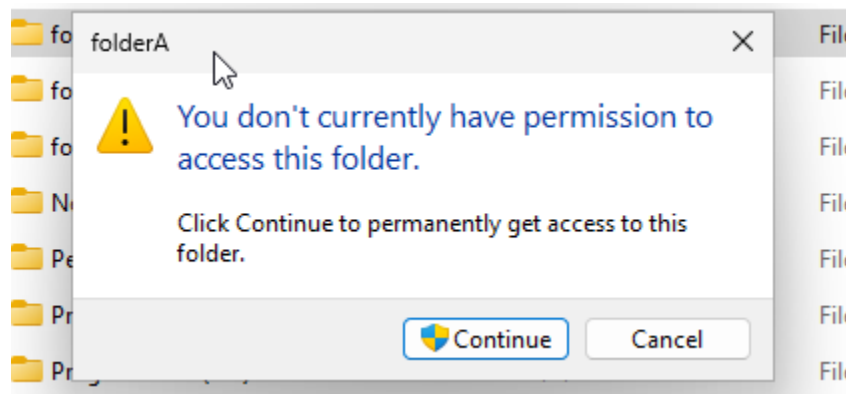




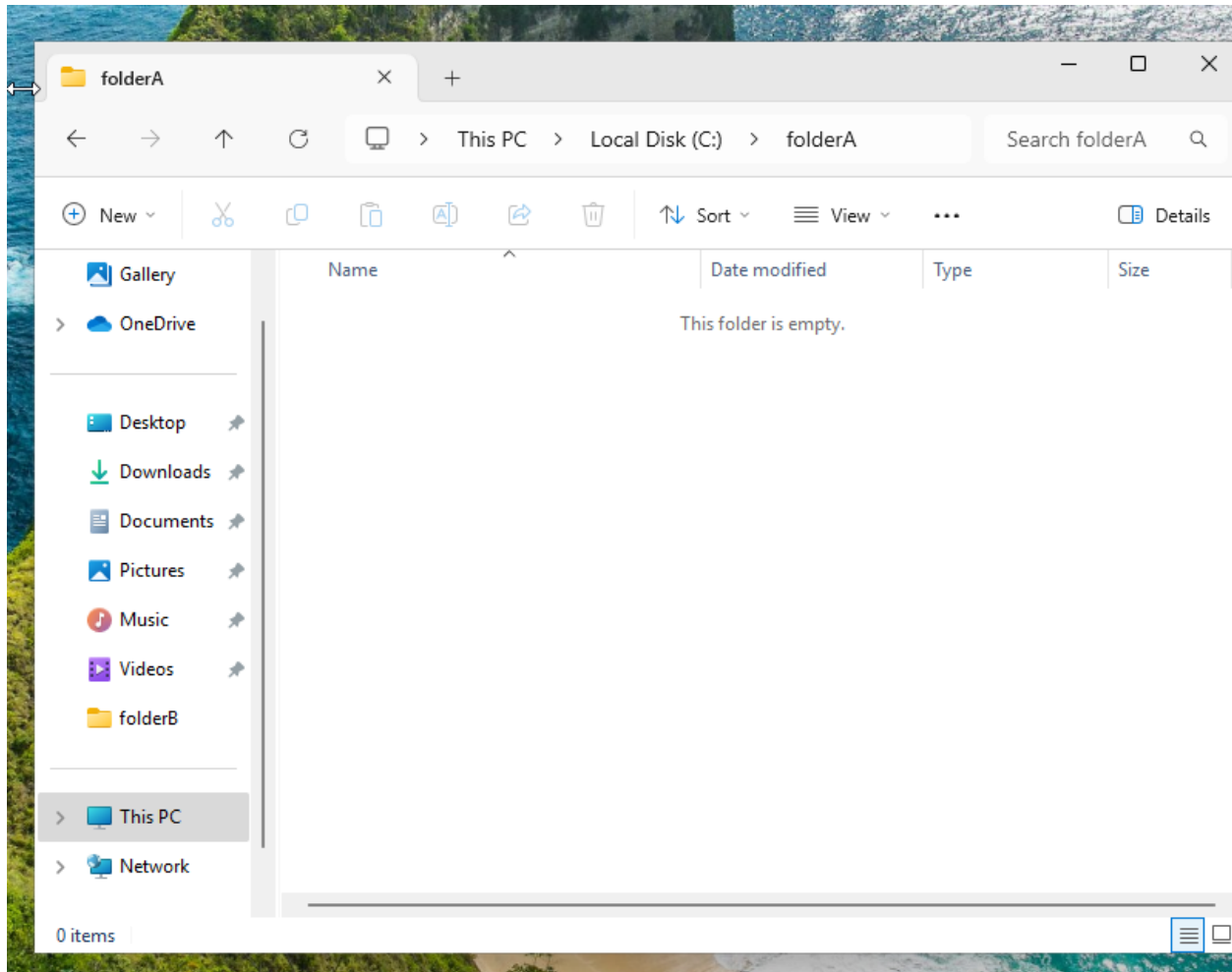
Next, while staying logged into my Windows 11 Client as the following account, I added the “CWeezer” user to the “IT” group.



Now if I try to access folder A again, I am still greeted with an error message.



However, if we log back in as the same user (CWeezer in my case), attempting to access the folder will now be successful.



This is due to when logging in with the “CWeezer” account, my login created an access token that all the groups a user is connected to. When I logged in originally, I was only a part of the “Advertising” group, and thus unable to open folder A. However, when I added the “CWeezer” user to the “IT” group and signed out and signed back in, a new token was created. Thanks to the updated access token, I was granted access to the file.

